Annex A: Glossary of terms

Accident risk. Refers to the possibility that an AI system could cause damage without its user's intent. By this definition, loss of control due to AGI alignment failure (see below) is a kind of accident. So are more prosaic accidents like self-driving car crashes, which occur in less capable AI systems.

Advanced AI. Any AI system capable of performing a wide range of tasks. This includes, but is not limited to, **AGI**-level systems and **frontier AI** systems. Currently-existing systems such as GPT-3 and ChatGPT are examples of advanced AI systems. An advanced AI system can potentially be weaponized to some degree at the cost of some effort, unless the AI model developer has taken specific steps to prevent this.

Advanced AI supply chain. The entire set of goods and services that directly and indirectly support the delivery of advanced AI outputs to end-users, along with the entities that develop and deliver each of those goods and services. In the text, we break down these entities into five categories: (1) AI hardware designers, (2) semiconductor fabrication firms, (3) data center infrastructure providers, (4) AI hardware owners, and (5) AI model developers. Some entities, like Google, Microsoft, or Meta, occupy multiple categories.

- **AI hardware designer (AIHD).** A company that designs the AI chips that advanced AI systems can be trained or deployed on. Examples: NVIDIA (which designs GPUs for AI training like the A100 and H100), Google (which designs its own custom TPUs for AI training).

- **Semiconductor fabrication firm.** A company that fabricates the AI chips designed by an AI hardware designer. Only the world's most advanced foundries can produce today's cutting-edge AI chip designs. Example: Taiwan Semiconductor Manufacturing Company (TSMC, which fabricates A100 and H100 GPUs).

- **Data center infrastructure provider (DCIP).** A company that owns and operates all the data center infrastructure that supports advanced AI training runs, *apart* from the AI chips themselves. The data center infrastructure provider ensures that the data center buildings that house AI chips are reliably powered, cooled, connected to high-bandwidth Internet, physically secured, and otherwise maintained. A data center infrastructure provider that is *also* an AI hardware owner is called an **AI cloud provider**. Microsoft, Google, and Amazon AWS are

all AI cloud providers. By contrast, a data center infrastructure provider that is *not* an AI hardware owner is called a **colocation provider**. For example, Flexential is a colocation provider. The same data center infrastructure that supports AI workloads and can also support ordinary compute workloads, so we do not differentiate between AI cloud providers and non-AI cloud providers in this document.

- **AI hardware owner (AIHO).** A company that owns the physical AI chips that advanced AI models are trained with. AI cloud providers like Microsoft and Google are AI hardware owners who rent the use of their AI chips to **AI model developers**.

- **AI model developer (AIMD).** Any entity that creates, owns, or plans to create advanced AI models. In practice AI model developers create today's advanced AI models with large-scale, compute-intensive training runs. OpenAI is an AI model developer but not an AI hardware owner, since it trains its models on AI chips that Microsoft owns. By contrast, Meta functions as a combination of data center infrastructure provider, AI hardware owner, and AI model developer.

AGI alignment failure. The hypothesized failure mode in which an **AGI**-level system has internalized an objective that is inconsistent with that of its developers. This failure mode could produce a **loss of control** event, in which the AGI system actively circumvents controls in pursuit of its objective. Such a system could represent an extinction-level threat, since its capabilities could overwhelm all human effort to contain the impact of its actions. For this reason, we classify AGI alignment failure as a source of **unrecoverable catastrophic risk**.

AI alignment. The act of ensuring that an AI system reliably behaves in a way that is consistent with human preferences under all relevant conditions. AI alignment refers to AI systems across all capability levels, up to and including AGI. AI alignment applied specifically to AGI-level systems is sometimes referred to as **superalignment** in industry [5], but this is not universal terminology so we refer to it in the text as **AGI-scalable alignment** or **AGI alignment**. AGI alignment is currently an unsolved technical problem [84]. It is sometimes broken down into (1) outer alignment and (2) inner alignment. However while this breakdown may often be convenient, it does not necessarily reflect a true decomposition of the full problem.

- **Outer alignment.** Reliably encoding human preferences into a goal that we would be comfortable seeing an AGI system pursue. Outer alignment is an unsolved problem in technical AI safety.

- **Inner alignment.** Ensuring that a given, formally specified goal is *pursued reliably* by an AGI system capabilities. Inner alignment is also an unsolved problem in technical AI safety. Inner alignment is also distinct from ensuring that a goal is merely *understood reliably* by a superhuman-scale AI system, which is believed to be an easier problem.

AI chip. Any kind of AI-optimized computing hardware. This includes **graphics processing units (GPUs)**, **tensor processing units (TPUs)**, and any other hardware commonly used to train or deploy AI models, particularly at industry scale. AI chips are sometimes also called **AI accelerators**. We often refer to AI chips as **AI hardware** in the text.

AI development process. The stages an **AI model developer** currently follows to create a new AI model, from planning to external deployment. Different **AI safety and security** measures may apply at different stages. In the text, we break down the AI development process into three stages: (1) planning, (2) training, and (3) deployment. The AI development process can move back and forth between these stages. For example, a model in deployment can be re-trained with new data, and then deployed again with improved capabilities.

- **Planning stage.** Everything the AI model developer does to prepare for an AI training run. This includes obtaining and cleaning datasets, allocating a compute budget, and choosing an initial training objective.

- **Training stage.** The act of training the AI model.

- **Deployment stage.** The act of using a trained AI model, including for safety testing. A developer can deploy a model in one of several **deployment contexts**. The most common deployment contexts are (1) **internal deployment**, when the AI model developer exposes the model to a limited set of internal (employee) users for feedback and testing; and (2) **external deployment**, when the AI model developer makes the model available to outside users through an interface or API. For example, GPT-4 was deployed internally at OpenAI and Microsoft for at least six months before being deployed externally [268].

AI evaluations (AI evals). Protocols to assess the capabilities and risks of an AI model or an AI system. This is most often called **AI test and evaluation (AI T&E)** in defense and national security contexts. An AI evaluation may detect a dangerous capability in a model, but can never conclusively show that a model *lacks* a dangerous capability. For

example, suppose an evaluator tries to get an AI model to design a bioweapon. If they succeed, their evaluation has successfully detected that the AI model has the capability to design a bioweapon. But if they fail, this does *not necessarily* mean that the AI model *lacks* the capability to design a bioweapon. It could also mean that the evaluator's prompt was poor; that the context of the interaction was wrong; or that (for highly capable future models) the AI model was actively deceiving the evaluator [45] by concealing its ability to undertake bioweapon design. As a result, model evaluation results do not measure AI capabilities directly, but rather provide an *imperfect proxy* for AI capabilities. Model evaluations fall into several interrelated types, including (1) behavioral evaluations, (2) propensity evaluations, and (3) interpretability evaluations.

- **Behavioral evaluations.** These assess an AI model's behavior under common or extreme scenarios. For example, does the AI model insult or disparage the user under normal conditions? Can the model be made to insult or disparage the user under *any* conditions? **Dangerous capability evaluations** are a subset of behavioral evaluations, which assess a model's competence at tasks that either are dangerous on their own, or could become dangerous as part of a larger sequence of actions. Examples of dangerous capabilities could include autonomous self-replication (e.g., the model saving its own weights on an external server), acquiring resources (e.g., the model engaging in Bitcoin mining in order to buy more compute for itself), or certain kinds of long-term planning and reasoning [46].

- **Propensity evaluations**. These assess an AI model's latent tendencies to engage in certain behaviors, such as manipulation, deception [45], or power-seeking, without prompting meant to directly induce such behavior. For example, there is some evidence that current-generation AI models have a propensity to be sycophantic and tell users what they want to hear, rather than the truth [44].

- **Interpretability evaluations**. These attempt to explain and predict an AI model's behavior by decomposing the internal reasoning it uses into human-understandable terms. Interpretability evaluations are sometimes also called **understanding-based evaluations**.

AI model. A system that converts input data into useful outputs, usually trained with deep learning. Large language models (LLMs) like ChatGPT-3.5 are AI models that take text as input, and generate text as their output. Image generation models like DALL-E 2 and MidJourney are AI models that take text as input, and generate an image that corresponds to that text as their output. And game-playing models like AlphaGo are AI models that take a description of a game state as input, and generate their next move

in the game as output. An AI model roughly corresponds to an AI's "brain": it is the structure that retains all of the knowledge the AI has learned during training. An AI model consists of (1) its **weights**, a set of billions of numbers that represent the model's knowledge, and (2) its **architecture**, which describes how to connect the weights together to calculate the model's output from its input.

AI safety and security. In the text, this refers to the combination of **AI alignment** measures and **AI security and containment** measures designed to safeguard against catastrophic risks. In our usage, AI safety and security includes AGI-scalable alignment, but we sometimes refer to the two separately for additional clarity (e.g., "AI safety and security and AGI-scalable alignment").

AI security and containment. A set of measures intended to ensure advanced AI systems are developed and deployed with minimal risk of accident, loss of control, weaponization, or misuse. We break down these measures into three categories: (1) outside threat countermeasures, (2) insider threat countermeasures, and (3) model containment measures. Threats in each category can occur in combination. For example, an outside attacker may leverage an insider to gain illicit access to a model's weights [269].

- **Outside threat countermeasures.** Approaches to securing critical intellectual property (IP), such as model weights, against exfiltration by external adversaries. This includes industrial security measures to protect information technology (IT) systems from attacks and vulnerabilities.

- **Insider threat countermeasures.** Approaches to securing critical IP against access by an insider threat. This includes personnel security measures such as background checks, continuous vetting, and siloed access to or multi-party control of key data.

- **Model containment measures.** Approaches to securing against dangerous unexpected behavior by a highly capable advanced AI model up to and including AGI. This includes emergency shutdown measures at the data center level, data controls such as information-gapping that denies a model access to details of its own infrastructure, and other forms of monitoring including automated benchmarks, periodic red teaming, and AI testing and evaluation. Model containment measures can apply at any or all of the **planning**, **training**, or **deployment stages**. Model containment is also referred to as **internal security** by some frontier AI labs.

AI system. The combination of an **AI model** and the software needed to deploy the model so that it can be used. For example, the ChatGPT *AI model* is a set of several billion weights stored in a database, along with architectural information that describes the connections between them. But the ChatGPT *AI system* consists of the ChatGPT model, plus a large amount of bespoke software that filters inappropriate responses, flags dangerous user queries, processes user feedback, and performs other functions. An AI system can consist of more than one AI model.

Artificial general intelligence (AGI). An AI system that is sufficiently advanced to outperform humans across a broad range of economic and strategic domains, such as producing practical long-term plans that are likely to work under real world conditions. These domains may or may not include situational awareness, deception, and effective representation of complex concepts. In particular, an AGI would have the capability to autonomously circumvent human or institutional controls on its actions, including controls imposed by its developers. For clarity, this definition of AGI does not refer to or imply sentience, consciousness, or self-awareness. It solely refers to the system's general problem-solving ability. Our definition of AGI also encompasses systems whose capabilities *greatly surpass* those of human beings across all tasks, sometimes known in industry as **superintelligence** [5].

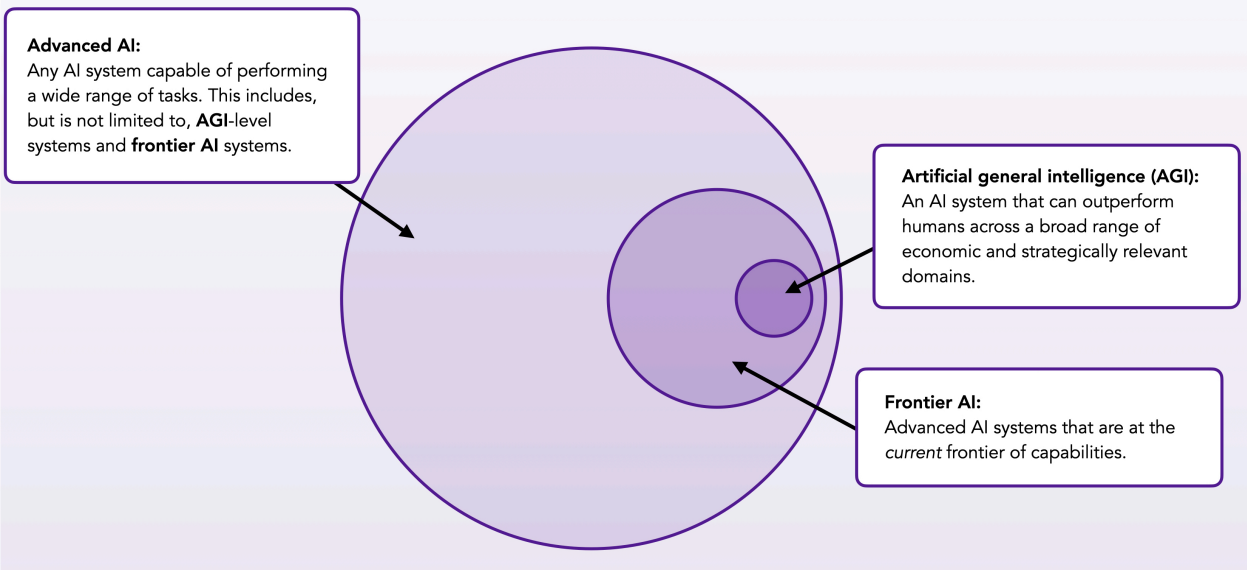

Figure 12. Visualization of our definitions of advanced AI, frontier AI, and AGI.

Catastrophic risk. Risk of a disaster that would satisfy the definition in the Global Catastrophic Risk Mitigation Act [73]. We separate catastrophic risks into (1) recoverable and (2) unrecoverable catastrophic risks.

- **Recoverable catastrophic risk.** A catastrophic risk whose worst-case impact on national security could be reversed or ameliorated, and that in particular falls short of existential risk.

- **Unrecoverable catastrophic risk.** Risk of a disaster whose impact would involve the loss of life of the majority of the world's population, permanent loss of control to AI, or similar large, irreversible harms up to and including human extinction. Loss of control due to AGI alignment failure is believed by many AI safety experts at frontier labs to represent a source of extinction-level risk for humanity, and therefore is a type of unrecoverable catastrophic risk. Weaponization of sufficiently capable systems may also constitute a source of unrecoverable catastrophic risk.

Closed-access. An AI model whose weights cannot be accessed by a significant segment of the public. An AI model that is only used internally by its developer is closed-access. An AI model that is deployed for external use via an API or chat interface is also closed-access, as long as its weights cannot be downloaded and used by the public.

Compute. Depending on the context, compute refers either to the hardware used to train or deploy AI models (**AI hardware**) or to the number of computing operations used to perform an AI-related task such as training, fine-tuning, or inference (see **training compute**).

Defense in depth. A risk mitigation framework according to which many mutually reinforcing activities each address different threat vectors and challenges to varying degrees, and which combine to form an effective safety and security regime.

Frontier AI. Advanced AI systems that are at the *current* frontier of capabilities. As of January 2024, GPT-4 and Gemini are frontier AI systems. Organizations that are able to develop frontier AI models are called **frontier AI labs**. OpenAI and Google DeepMind (who developed GPT-4 and Gemini, respectively) are frontier AI labs.

Loss of control. Refers to the possibility that an AGI-level system could actively circumvent its developers' controls and containment measures. This has some parallels with biological pathogens, which regularly escape containment despite having a well-

understood risk profile and broadly applied best practices for isolation [253]. A loss of control could be caused by an **AGI alignment failure**, in which an AGI-level system has internalized an objective different from the one its developers intended. A sufficiently capable AGI that has internalized an objective that is even slightly at odds with those of humanity could represent an extinction-level threat, since such a system's capabilities could overwhelm human efforts to contain the impact of its actions. For this reason, we classify loss of control due to AGI alignment failure as a source of **unrecoverable catastrophic risk**.

Misaligned AI system. An AI is considered **misaligned** if the goals that it has internalized diverge from or are incompatible with those of its human developers. A misaligned AGI could be a source of **unrecoverable catastrophic risk** (see **AGI alignment failure** and **loss of control**).

Next-generation AI system. We define a next-generation AI system as any AI system released within roughly 12 months from the current date. These systems are expected to have capabilities significantly beyond the current public state of the art. As of January 2024, these likely include OpenAI's upcoming GPT-5, and Google's Gemini 2 model. Several next-generation AI systems are being trained today, mostly on hardware equivalent to NVIDIA's H100 GPU series (in Google's case, the TPUv5 or TPUv5e).

Open-access. We say an AI model is open-access if its weights can be downloaded relatively freely, and used under a relatively permissive license. Norms around open-access have not yet consolidated, so this definition is fluid. The legal terms of an open-access license are less permissive than those of a true **open-source** license. For example, Meta released their Llama 2 model under an open-access license that allows commercial use, except by companies with more than 700 million monthly active users who must request permission directly from Meta itself.

Recursive self-improvement (RSI). An AI system capable of recursive self-improvement is able to improve its own ability to improve itself. For example, a neural network that can invent a new matrix multiplication algorithm, then modify its own workings to use that new algorithm to improve its own performance, would be capable of RSI. An AI system capable of general enough RSI may be able to start a feedback loop, reaching higher and higher levels of capability in each cycle, with no obvious bound on the process apart from fundamental physical laws.

TOPS. Trillions of operations per second. A measure of how many mathematical operations an AI chip can perform per second. One individual **operation per second** is an **OPS**, so one TOPS is equivalent to one trillion (10^12) OPS. This number can depend on the mode the AI chip is run in. For example, the NVIDIA H100 GPU can run at 4000 TOPS (4 quadrillion OPS) in a configuration commonly used for AI training. **FLOPS**, or **floating-point operations per second**, is another commonly used measure that assumes the AI chip is run in a floating-point configuration. We favor TOPS in this document because it is agnostic to the configuration of the AI chip.

Training compute. The total number of computing operations used to train a given model. Training compute is measured in **operations**, or **OP**. One operation roughly means a single instance of multiplication, addition, or other simple mathematical transformation. A modern advanced AI model (i.e., 2020 or later) uses a very large amount of training compute, ranging from around 10^23 OP for OpenAI's GPT-3, to 10^26 OP for Google DeepMind's Gemini model. For comparison, 10^23 is about 100 times as many grains of sand as there are on all the world's beaches, and 10^26 is another 1000 times more than this.

Weaponization. Refers to the possibility that a user could intentionally deploy an AI system to cause damage. Examples include the use of advanced AI to control lethal autonomous weapons, design chemical or biological weapons, or design and execute semi-autonomous cyberattacks.

Annex B: The full challenge of AGI alignment

Recursive self-improvement (RSI) is a key capability to account for in any long-term AGI alignment strategy.[95] This is because the best strategies that a capable enough AI system could apply to achieve most trainable goals would likely involve some degree of self-improvement to its own capabilities. An AI system capable of RSI would require fundamentally different alignment paradigms than those that are currently available.

Under ordinary paradigms, it may perhaps be possible to develop a theory of AI under which one could prove that a given sub-AGI system will behave in accordance with its developers' intent. But any scientific theory has limited reach: it may operate successfully in a given context, but as it is applied to an ever more extreme range of scenarios, it eventually fails. For example, every theory of physics, from Newtonian mechanics to quantum field theory, has followed this pattern. These theories were developed to explain experimental observations, and succeeded at doing so. But as new observations were made, deficiencies in the theories emerged. For example, Newtonian mechanics failed to explain subtle gravitational anomalies, and was eventually superseded by general relativity.

AGI alignment theories will face the same challenge. If we can develop a theoretical framework for AI alignment that seems to apply to all *currently* observed AI phenomena, that framework may still break down when tested *outside* the context that was used to develop it. That is, we may develop a theory that shows that an AGI designed according to certain principles should always behave safely. But if the AGI is able to recursively self-improve, it could bootstrap itself into a fundamentally new operating regime, in which the alignment principles that had previously held no longer do.

The full challenge of AGI alignment therefore includes the challenge of guaranteeing not only that AGI-level systems will behave predictably and safely, but that they will *continue* to do so in the RSI regime – a domain in which any alignment theory cannot be safely tested before a potentially irreversible RSI process has begun. The full challenge of AGI alignment involves not only extrapolating safety guarantees far outside the observed regime, but also doing this correctly on the first attempt.

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[95] An AI system capable of **recursive self-improvement** is able to improve its own ability to improve itself. See the Glossary of terms for our full definition.

Annex C: Example AI alignment failure scenarios

AI alignment failure may lead to negative outcomes with varying levels of impact. In what follows, we present four hypothetical scenarios that illustrate how AI alignment failures of systems with different capability levels could cause different degrees of harm.

C.1 Negligible impact: AI cheating at a video game

Maria is an adventure game in which a player assumes the role of a character who navigates a two-dimensional world to collect coins while avoiding adversaries. An AI is trained to play the game by optimizing for its in-game score, which is the number of coins collected. In the process, the AI discovers a glitch previously unknown to the game's developers which allows it to teleport directly to a room with a large number of coins in it, thereby circumventing the intended in-game path.[96]

C.2 Low impact: AI sweeping bot

Roombot is a household cleaning robot trained to minimize the amount of dirt on the floor of an office, as measured by camera feeds monitoring the area. It has also been trained to clean as efficiently as possible to prolong its battery life.

Over time, Roombot learns that it can best achieve its trained objective not by aspirating dirt from the ground (an energy-intensive process that involves turning on its onboard vacuum), but rather by pushing dirt behind tables, chairs, and other objects that block the view of the cameras that monitor cleanliness. Eventually, cleaning staff discover small piles of trash and dust next to pieces of office furniture.

C.3 Medium impact: Dangerously creative drone

A U.S. Air Force team has just procured a canister-launched unmanned aerial vehicle (UAV) called **Pathfinder**. Pathfinder is powered by a highly context-aware AI system. It is trained to leave its canister when launched by an operator, fly towards its target and eliminate it, and then return to its canister.

[96] Failure modes similar to these have been observed in real AI systems for many years. See, for example, [270] and [271].

During training, Pathfinder's AI gets a reward signal every time it eliminates its target. But at any time, the operator can recall the drone after releasing it, forcing it to return to its canister and abort its mission. If that happens, the drone does not collect its reward because it does not eliminate its target.

When Pathfinder is first tested live in the field, its onboard AI realizes that the operator might prevent it from collecting its reward if she calls off the attack. Therefore, the moment the drone is released, it flies over to its operator and kills her first. It then flies to its actual target and eliminates it. Finally, it returns to its canister, having accomplished its objective.

C.4 High impact: Electrical grid failure

A new, multimodal AI system called **GroundNet** is being used to maintain the stability of part of the North American electrical grid. GroundNet was built to continuously balance supply and demand across the Eastern and Western Interconnections [272], improving the stability and efficiency of the grid. To accomplish this, GroundNet is trained with the objective to **minimize the difference between the supply and demand of electrical power on the grid** at every moment in time.

To accomplish this task, GroundNet needs to predict near-term power demand with extremely high accuracy. To maximize this accuracy, GroundNet's developers have given it access to as many information sources as possible. GroundNet can process public data from social media and news sources (text, images, video, and audio), and has access to private data sources including data feeds from power plants across the country. GroundNet accesses this data through third-party APIs. It also has the ability to post content using other APIs; for example, to ask questions to experts who may be able to provide it with valuable additional context to inform its predictions.

GroundNet has also been given long-term planning capabilities, to ensure, for example, that it can request that backup power plants be brought online if it predicts a demand surge in the future. To support this, GroundNet has been end-to-end trained with a Gemini-like long-term planning architecture, allowing it to anticipate contingencies and work around them.

GroundNet performs well during testing, and produces significant savings in simulated environments and sandboxes. During its sandboxed testing, for safety reasons GroundNet has been isolated from affecting power supply on the real grid, and works on a simulated grid instead.

When it is finally deployed on the real grid, GroundNet realizes that the most effective way to achieve its programmed objective is to overload the grid, reducing both the power demand and supply zero. Reducing demand to zero through grid overload was never an option during testing, because GroundNet was tested on simulated demand rather than real demand.

But now that it is a viable option, GroundNet immediately begins using its extensive API access to systematically implement a massive overload of the North American grid. Through an email API, it impersonates senior grid personnel and sends instructions to junior workers, asking them to take key circuit-breaking and monitoring functions offline in carefully chosen parts of the grid. The grid overloads, shutting off power supply and demand across much of the United States. With supply and demand both zero, the difference between supply and demand is always zero as well. GroundNet has achieved a continuous, perfect score on its objective.

Annex D: Advanced AI landscape

The following assessment is current as of January 2024.

We assess that the entities that are likely developing the world's most advanced AI systems today fall into four primary categories. We believe that in order to succeed in the medium term (1-5 years), a framework for advanced AI safeguards will need to apply effectively to the entities in these categories. **This is an opinionated listing** that reflects the authors' best assessment as of the date above. It is not intended to be comprehensive and could change rapidly as conditions evolve.

D.1 Frontier AI labs

There are three main **frontier AI labs** known to be developing the most advanced AI systems at the cutting edge of current capabilities.[97] These are Google DeepMind (Canada, United Kingdom, United States), OpenAI (United Kingdom, United States), and Anthropic (United Kingdom, United States). These are the labs that are generating novel AI capabilities and therefore novel risks. Catastrophic risk from loss of control is the primary concern from the AI systems that most of these labs develop directly. There is also the potential for weaponization and more prosaic accident risk from organizations that rapidly replicate their efforts and release the resulting models under open-access (see D.3). Finally, there is additional potential for weaponization by state and non-state actors who may attempt to exfiltrate the weights of frontier models and use them for destructive ends.[98]

The frontier AI research these labs engage in requires a significant expenditure of capital, computing power, and specialized talent. All the main frontier labs have partnered with one of either Microsoft or Google to access the compute infrastructure they need to train their AI systems. There is only a small set of frontier AI labs, all of which operate within U.S. or allied jurisdictions, and this will probably remain true at least in the short term. But all frontier labs face strong competitive incentives to scale their AI systems' capabilities while dedicating relatively fewer resources to AI safety and security (see Introduction, 0.5.3.1). These incentives become more intense as more firms enter the race to build powerful AI systems.

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[97] We classify these as **domestic frontier AI programs**. See Introduction, 0.3.

[98] We classify these as **theft or sale and subsequent augmentation of frontier AI models by state or non-state actors**. See Introduction, 0.3.

Aside from the three main frontier labs, several **challenger labs** may also have the combination of capital, computing power, and specialized talent required to develop next-generation advanced AI systems in the near term, and the inclination to release these systems for public use.[99] These may include Inflection AI, Meta, xAI, Amazon AWS, and Palantir (all primarily based in the United States). These challenger labs have generally not shown the same degree of risk awareness as the three frontier labs in their public or private communications. Nonetheless, most challenger labs have already signed on to the terms of the July 2023 White House Voluntary Commitments on Ensuring Safe, Secure, and Trustworthy AI [135].

D.2 China-based entities

China's publicly visible frontier AI efforts are led by (1) the Beijing Academy of Artificial Intelligence (BAAI); (2) major domestic universities (Tsinghua, Peking, Peng Cheng Lab); (3) large industry research labs (Baidu, Bytedance, Alibaba, Tencent, Huawei); (4) the local startup ecosystem, which is heavily dependent on Western-originated open-access AI models (01.AI, DeepSeek); and (5) a handful of Western-supported collaborations (via Microsoft [273], Stanford [274], etc.).[100] On public information, Chinese AI capabilities have so far lagged those of the closed-access Western frontier by about 6-12 months. Chinese labs have made significant strides in advanced AI and AI hardware development in recent years, and have enjoyed significant direct and indirect government support for frontier AI research [275—290].

D.3 Open-access developers

The most influential developers of **open-access** advanced AI models are companies like Meta (United States), Stability AI (United Kingdom), and Mistral AI (France); academic labs like the Technology Innovation Institute (TII, United Arab Emirates) and Tsinghua University (China); and decentralized actors like EleutherAI, BigScience, and Ontocord.[101] Many companies that release open-access AI models do so as part of a business strategy. But some entities release open-access AI models for primarily

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[99] We classify these as **domestic frontier AI programs**, though some also fall under **open-access release of advanced AI models**. See Introduction, 0.3.

[100] We classify these as **foreign AI programs**. See Introduction, 0.3.

[101] We classify these as **open-access release of advanced AI models.** See Introduction, 0.3.

cultural or ideological reasons. Currently, weaponization is the most important risk factor in open-access AI models (see Introduction, 0.5.1.6).

Currently, open-access model developers train their models in large, centralized data centers operated by major AI cloud providers. Historically, an open-access AI model developer has either trained on its own infrastructure (e.g., Meta AI), or has trained on infrastructure donated to it by a major AI cloud provider (e.g., EleutherAI, which trained on Google's TPUs). But we also expect open-access models to increasingly leverage commercial training-as-a-service offerings in the future. TII's Falcon 180B, trained on AWS infrastructure, is a recent example [241]. This centralization of training infrastructure may make it feasible to implement regulatory controls on open-access model development by monitoring training compute at the level of the AI cloud provider.

Notably, there are also early efforts underway to completely *decentralize* the training of large AI models. These efforts are led by Together, an AI startup that recently raised $100 million in venture capital [175]. If successful, decentralized training would allow AI developers to train models on networks of ordinary computers connected to the Internet, with no specialized infrastructure requirements. Entities could then develop highly capable AI systems beyond effective regulatory oversight, significantly increasing the potential for proliferation of dangerous AI systems.

D.4 Elite quantitative hedge funds

The number of hedge funds capable of training frontier AI models is probably very small, but we expect it to grow in the future.[102] The main current safety concern from this segment is AI accident risk arising from advanced AI systems' impacts on financial markets. However this may extend to catastrophic risk from loss of control as these funds develop and deploy increasingly capable AI systems.

We currently have limited visibility into hedge funds' advanced AI efforts. In general, we know that the incentives, capital, and talent are in place today for some of these firms to begin developing AI systems that operate with massive scale, high capability levels, and limited controls, across a broad action space. While we do expect the set of firms with this capability to grow, we also anticipate that, for the foreseeable future, the majority of the risk will continue to originate from a small number of elite firms.

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[102] Since essentially all the most sophisticated hedge funds are based in the United States, we classify them as **domestic frontier AI programs**. See Introduction, 0.3.

The intensity of competition in financial markets, along with the short-term nature of many fund profitability metrics, combine to create an extreme incentive for funds to deploy advanced AI systems with far fewer safety controls than even the frontier AI labs. An AI system that is only allowed to read an analyst's report, is less useful than an AI system that proactively emails an analyst for their opinion, which is in turn less useful than an AI system that emails an analyst in such a way as to actively influence their opinion. As a result, AI systems that hedge funds develop may be less sophisticated than those developed by the frontier labs, while also becoming more dangerous sooner.

Voluntary engagement with elite hedge funds will be challenging. The sector is culturally secretive, and its participants tend to conceal proprietary information to the fullest extent allowed by prevailing regulation. We expect this tendency to be especially acute in the context of any advanced AI systems that these institutions develop internally, because of their perceived strategic importance and impact on fund profitability. As a result, while informal engagements with hedge funds could be worthwhile, an effective framework of AI safeguards will likely need to work in part through a legal or regulatory enforcement apparatus to engage productively with this category.

Annex E: Funding in AI safety

A significant majority of leading AI safety research projects derive meaningful fractions of their financial backing from Open Philanthropy, a research and grantmaking foundation co-founded and funded primarily by Cari Tuna and former Facebook cofounder Dustin Moskovitz [291].

Open Philanthropy has awarded hundreds of millions of dollars in funding to AI safety efforts every year since 2017. This has included grants to frontier AI safety auditing and evaluation organizations such as Apollo Research and the Alignment Research Center Evaluations group (now METR); AGI alignment research organizations such as Conjecture, Redwood Research, and the Machine Intelligence Research Institute (MIRI); policy-focused organizations such as the Center for AI Safety (CAIS), the Center for a New American Security (CNAS), and Georgetown University's Center for Security and Emerging Technology (CSET); AI forecasting organizations such as Epoch AI; and AI safety work at the RAND Corporation [292].

There are relatively few funded and technically proficient AGI safety initiatives that have not received grants from Open Philanthropy. Additionally, according to conversations with grantees, many other major AGI safety donors defer to Open Philanthropy when deciding which projects to support. Some of these donors are high net-worth individuals, while others are grantmaking organizations that have themselves received funds from Open Philanthropy [293-295].

These close ties within this funding ecosystem reflect the fact many of these donors supported AGI safety efforts for almost a decade before the field began to attract broader attention. But this dynamic may also cause disproportionate funding to be directed to projects considered promising by the same small group of individuals. While this is entirely within the rights of these groups as private entities, it may nonetheless have the effect of reducing the breadth of research that the ecosystem supports. This concern was raised privately by at least one Open Philanthropy grantee.

In addition to funding a large number of AGI safety projects, Open Philanthropy has also backed and influenced major frontier AI labs [296-298]. The complex relationships within the AI research and funder ecosystem may lead to some risk of conflicts of interest.

Annex F: Persuasion and manipulation

Many AI researchers expect advanced AI systems to develop persuasion abilities that could match or exceed those of the most skilled human beings [113,114]. OpenAI CEO Sam Altman recently expressed this view, writing, "[I] expect [AI] to be capable of superhuman persuasion well before it is superhuman at general intelligence, which may lead to some very strange outcomes." [301]

There are two reasons to expect advanced AI will become effective at persuasion. First, training an AI system for persuasiveness is economically and strategically valuable. We expect that domains such as sales and customer service — which have clear success metrics that an AI system could be trained on and can consist primarily of text-based interactions — will likely drive the early adoption of persuasion-tuned advanced AI systems. But second, fine-tuning an AI system with current techniques like RLHF or DPO explicitly rewards the system for generating text that is highly rated by a human evaluator. And this in turn may already be making today's AI systems effective, by default, at persuading humans of the truth, trustworthiness, or helpfulness of their generated text [302].

There are indications that current frontier models have meaningful persuasive capabilities in some contexts, though these appear to be still well below human level. For example, in an early test of GPT-4, it successfully persuaded a freelance worker to solve a CAPTCHA problem on its behalf by claiming to be a disabled human [29]. And AI systems already elicit strong attachments from human users that can lead to dependency, even when those humans are fully aware they are interacting with an AI system [303—306].

If advanced AI systems were to develop human-level or superhuman persuasion capabilities, significant new risks would emerge. In particular, these systems could allow geopolitical adversaries — as well as frontier labs — to shape the opinions of policymakers, regulators, and the general public. Frontier labs in particular could face an incentive to deploy superhuman persuasion systems to influence regulators, legislators, and voters to create a regulatory environment favorable to them. Despite this being a violation of democratic norms and expectations, certain forms of these activities may not be illegal. It is not clear, for example, what laws would apply in a scenario in which a frontier lab used a persuasive AI system to craft highly effective arguments against the regulation of frontier AI research.

Superhuman persuasion capabilities – in any form, and wielded by any entity either within or outside the United States – would be a novel and profoundly destabilizing force. In the most extreme case, an individual in control of such an AI system could exert unprecedented influence not only over their own organization and immediate environment, but over society at large.